Protocol

Biometeorological indicators for child and adolescent health, cognition and neurodevelopment: a scoping review

29 Oct 2026

Administrative information

This is a scoping review of how the thermal environment has been characterised in studies of children's and adolescents' health, cognition and neurodevelopment. Its applied purpose is to justify the choice of exposure indicators for the school-level heat exposure assessment of EscuelasCool (WP2.1).

Item	Detail
Working title	Biometeorological indicators for child and adolescent health, cognition and neurodevelopment: a scoping review
Review type	Scoping review (JBI methodology), with a pre-specified analytical subset of comparative-performance studies
Reporting	PRISMA-ScR; protocol following JBI guidance for scoping review protocols
Registration	Github-Zenodo
Project	EscuelasCool — Making a Silent Killer Visible (WP2.1)
Funder	Wellcome Trust Climate Impacts Award, ref. 331072/Z/25/Z

Roles follow CRediT. The funder has no role in design, conduct, interpretation or reporting.

Background and rationale

The exposure metric chosen to represent heat shapes the estimated exposure–response relationship, the thresholds that can be proposed for alerts and school guidance, and the projected burden under future climate. More than a hundred human thermal indices exist, ranging from single meteorological variables to empirical combinations (e.g. Heat Index, Apparent Temperature, Humidex, Wet-Bulb Globe Temperature) and heat-balance models (e.g. PET, UTCI) (de Freitas and Grigorieva, 2015).

Existing syntheses are dominated by adults and by mortality

The most comprehensive systematic review to date included 310 studies and 1,143 associations, and identified 17 indices, with Apparent Temperature the most frequently used (Pantavou et al., 2025). About half of the associations concerned mortality; infants, children and adolescents accounted for only 74 morbidity associations (13.6%), and tropical climates for 4.1% of all associations. That review excluded indoor environments and physiological outcomes, and methodological heterogeneity precluded meta-analysis. A scoping review on maternal and perinatal health identified four indices and

called for their standardisation and explicit justification (Brimicombe et al., 2024). No equivalent synthesis exists for children and adolescents.

Comparative evidence is inconclusive

In some settings, complex indices did not outperform air temperature in predicting heat-related mortality (Morabito et al., 2014; Vaneckova et al., 2011), and in older US adults the choice between temperature and Heat Index did not substantively change risk estimates (Spangler et al., 2023). Conversely, pediatrics studies have reported that UTCI or WBGT captured effects not visible with air temperature alone (UTCI and pediatrics admissions, Pediatrics 2025 [https://pubmed.ncbi.nlm.nih.gov/39729086/]; WBGT and children's emergency visits in Taiwan [https://www.sciencedirect.com/science/article/pii/S0048969724069973]). Whether these findings hold for school-age children, for non-fatal and cognitive outcomes, and in tropical and Andean settings is unknown.

Children raise specific measurement problems

Pediatric thermoregulation differs from adult thermoregulation in body surface-to-mass ratio, sweating and the relative reliance on dry heat exchange, although the current evidence is nuanced. Children's exposure is also structured by context: compulsory school attendance, fixed timetables, uniforms, outdoor recess and physical education, and little control over classroom conditions. Most heat-balance indices assume an adult reference person, and most outdoor indices do not represent the classroom. Reviews of heat and learning have, moreover, relied largely on air temperature (Vasilakopoulou and Santamouris, 2025).

Link to EscuelasCool WP2.1.

WP2.1 will characterize historical and projected (2000–2050) heat exposure for approximately 177,000 schools in Chile, Colombia and Peru, using ERA5-Land and CMIP6, restricted to school hours (07:00–17:00 local time). A small set of indicators must be selected that is physiologically meaningful for children, supported by pediatric evidence, computable consistently from gridded historical and projected data, and usable for policy thresholds. This review provides the evidence base for that choice.

Objectives and review questions

Primary objective

To map and characterize the biometeorological metrics used to study the effects of the thermal environment on the health, cognition and neurodevelopment of children and adolescents, and to synthesize the evidence on their comparative performance, in order to inform the selection of school-level exposure indicators.

Review questions

1. **Metrics (RQ1).** Which thermal metrics have been used, and how are they defined? This covers metric class (single variable, empirical index, heat-balance index, event-based metric), input variables, temporal aggregation (hourly, daily mean/maximum/minimum, school hours, cumulative or seasonal), and threshold definition (absolute, relative percentile, index-specific stress categories).

2. **Exposure assessment (RQ2).** How was exposure obtained and assigned? This covers data source (station, gridded reanalysis, satellite, modelled, measured indoor or outdoor), spatial resolution, assignment unit (residence, school, municipality) and handling of indoor versus outdoor exposure.
3. **Comparative performance (RQ3).** Which studies compared two or more metrics within the same population, using which criteria (model fit, predictive accuracy, effect size, threshold detection), and with what results?
4. **Distribution of the evidence (RQ4).** How is the use of metrics distributed by age group, outcome domain, study design, Köppen climate and region, with particular attention to Latin America?
5. **Pediatric justification (RQ5).** What child-specific reasons, if any, do authors give for their choice of metric (physiology, activity, clothing or uniforms, time-activity patterns, indoor exposure)?

Eligibility criteria

Studies are eligible if they quantitatively relate a thermal metric measured or estimated during childhood or adolescence to a health, cognitive or neurodevelopmental outcome in the same age group.

Element	Include	Exclude
Population	Children and adolescents aged 0–19 years (WHO definition of adolescence). Mixed-age studies only if they report estimates specific to ≤19 years	Adult-only populations; mixed-age studies without age-specific estimates
Exposure timing	Postnatal exposure during childhood or adolescence	Prenatal-only exposure (covered by existing maternal and perinatal reviews)
Concept: exposure	Any thermal metric across the full temperature range (heat and cold): air temperature, humidity-integrated indices, radiation- and wind-integrated indices, heat-balance indices, event metrics (heatwaves, cold spells, tropical nights), indoor or outdoor	Studies where temperature is only a covariate and no estimate for it is reported
Concept: outcomes	Mortality; morbidity (emergency visits, admissions, heat-related illness, disease-specific outcomes); physiological strain (core or skin temperature, heart rate, hydration); cognitive and academic performance; attendance; neurodevelopment; mental health and behavior; sleep	Thermal sensation or comfort votes as the only outcome (charted separately if they validate an index against children's physiological responses)
Context	Any country and climate; any setting (school,	—

Element	Include	Exclude
	home, community, sport)	
Study design	Observational (time series, case-crossover, cohort, panel, cross-sectional, case-control); experimental and quasi-experimental; methodological studies comparing metrics	Reviews (used for citation chasing only); projection studies that apply external exposure–response functions without estimating one; editorials, commentaries
Language	English, Spanish, Portuguese	Other languages (listed in an appendix)
Dates	Database inception to search date	—

Search strategy

The search uses existing reviews as a sampling frame and a benchmark, then searches primary studies directly so that metrics, cold exposure and indoor settings not covered by earlier reviews are captured.

1. **Stage A — reviews as sampling frame.** Identify systematic, scoping and umbrella reviews and meta-analyses on the thermal environment and the health, cognition or learning of children and adolescents. Extract all their included primary studies into a *seed set*. The seed set is used both as a source of studies and as a benchmark to test the sensitivity of the Stage B search.
2. **Stage B — primary studies.** Search PubMed/MEDLINE, Web of Science Core Collection, Scopus and LILACS (BVS/BIREME) from inception. Strings combine three concept blocks (population, thermal exposure, outcome), with controlled vocabulary where available.
3. **Stage C — citation chasing.** Backward and forward citation searching of all included studies and of key reviews, using the `citationchaser` R package (Lens.org).
4. **Grey literature.** Education-economics working papers, where much of the heat–learning evidence appears (NBER, IZA, World Bank and IDB working paper series, RePEc/IDEAS), and reports from WHO, UNICEF and national meteorological services in the region.

Draft PubMed string (to be checked and tested)

```
#1                                                                    Population
("Child"[Mesh] OR "Adolescent"[Mesh] OR "Infant"[Mesh] OR "Students"[Mesh] OR
"Schools"[Mesh]
 OR child*[tiab] OR adolescen*[tiab] OR paediatric*[tiab] OR pediatric*[tiab] OR
infant*[tiab]
 OR toddler*[tiab] OR teen*[tiab] OR youth*[tiab] OR "young people"[tiab] OR
schoolchild*[tiab]
 OR pupil*[tiab] OR student*[tiab] OR school*[tiab] OR kindergarten*[tiab] OR
preschool*[tiab])

#2                                  Thermal                                 exposure
```

```
("Hot Temperature"[Mesh] OR "Cold Temperature"[Mesh] OR "Extreme Heat"[Mesh] OR "Humidity"[Mesh]
 OR "ambient temperature*"[tiab] OR "air temperature*"[tiab] OR "outdoor temperature*"[tiab]
 OR "indoor temperature*"[tiab] OR "classroom temperature*"[tiab] OR heatwave*[tiab] OR "heat wave*"[tiab]
 OR "extreme heat"[tiab] OR "high temperature*"[tiab] OR "cold spell*"[tiab] OR "heat stress"[tiab]
 OR "thermal stress"[tiab] OR "thermal environment*"[tiab] OR "thermal index"[tiab] OR "thermal indices"[tiab]
 OR biometeorolog*[tiab] OR "heat index"[tiab] OR "apparent temperature"[tiab] OR humidex[tiab]
 OR "wet bulb"[tiab] OR "wet-bulb"[tiab] OR WBGT[tiab] OR UTCI[tiab] OR "universal thermal climate index"[tiab]
 OR "physiological equivalent temperature"[tiab] OR "effective temperature"[tiab]
 OR "perceived temperature"[tiab] OR "discomfort index"[tiab] OR "wind chill"[tiab]
 OR "tropical night*"[tiab] OR "diurnal temperature range"[tiab])

#3 Outcomes
(mortality[tiab] OR death*[tiab] OR morbidity[tiab] OR hospital*[tiab] OR "emergency department*"[tiab]
 OR admission*[tiab] OR "heat-related illness*"[tiab] OR "heat illness*"[tiab] OR health[tiab]
 OR cogniti*[tiab] OR "academic performance"[tiab] OR "academic achievement"[tiab] OR "test score*"[tiab]
 OR learning[tiab] OR attention[tiab] OR "reaction time"[tiab] OR neurodevelopment*[tiab]
 OR "mental health"[tiab] OR behavio*[tiab] OR sleep[tiab] OR attendance[tiab] OR absenteeism[tiab]
 OR "heart rate"[tiab] OR "core temperature"[tiab] OR dehydrat*[tiab])

#4 Exclusions (to be tested for recall loss)
("Hypothermia, Induced"[Mesh] OR "Fever"[Majr] OR incubator*[ti] OR burn*[ti])
 OR ("Animals"[Mesh] NOT "Humans"[Mesh])

#1 AND #2 AND #3 NOT #4
#1 AND #2 AND #3 NOT #4
```

Draft Web of Science Core Collection string (check and test)

```
TS=(child* OR adolescen* OR paediatric* OR pediatric* OR infant* OR toddler* OR teen* OR youth*
 OR "young people" OR schoolchild* OR pupil* OR student* OR school* OR kindergarten* OR preschool*)

AND TS=("ambient temperature*" OR "air temperature*" OR "outdoor temperature*" OR "indoor temperature*"
 OR "classroom temperature*" OR heatwave* OR "heat wave*" OR "extreme heat" OR "high temperature*"
 OR "cold spell*" OR "heat stress" OR "thermal stress" OR "thermal environment*" OR "thermal ind*"
 OR biometeorolog* OR "heat index" OR "apparent temperature" OR humidex OR "wet bulb" OR "wet-bulb"
 OR WBGT OR UTCI OR "universal thermal climate index" OR "physiological equivalent temperature"
```

```
 OR "effective temperature" OR "perceived temperature" OR "discomfort index" OR "wind
chill"
 OR      "tropical      night*"      OR      "diurnal      temperature      range")

AND TS=(mortality OR death* OR morbidity OR hospital* OR "emergency department*" OR
admission*
 OR "heat-related illness*" OR "heat illness*" OR health OR cogniti* OR "academic
performance"
 OR "academic achievement" OR "test score*" OR learning OR attention OR "reaction
time"
 OR neurodevelopment* OR "mental health" OR behavio* OR sleep OR attendance OR
absenteeism
 OR      "heart      rate"      OR      "core      temperature"      OR      dehydrat*)

NOT TS=("induced hypothermia" OR "therapeutic hypothermia" OR incubator* OR febrile)
NOT TS=("induced hypothermia" OR "therapeutic hypothermia" OR incubator* OR febrile)
```

Strings for Scopus and LILACS (with DeCS terms in Spanish and Portuguese) will be translated from these and reported in full in the supplementary material. University students retrieved by `student*` are excluded at screening.

Evidence selection

Records are deduplicated in R and screened in Rayyan in two stages: titles and abstracts, then full texts.

- **Title and abstract screening.** One reviewer screens all records and a second screens a random 20% sample; if agreement falls below 80%, dual screening is extended to all records. Records are only excluded at this stage when ineligibility is clear.
- **Full-text screening.** Two reviewers independently assess all full texts. Disagreements are resolved by discussion or by a third reviewer. Reasons for exclusion are recorded and reported in the PRISMA-ScR flow diagram.
- **Overlap with the companion review.** Both reviews share deduplicated records. Studies with cognitive, academic or neurodevelopmental outcomes that meet both sets of criteria are tagged once and charted with a common form (see *Data charting*), so that each review uses the fields it needs without duplicated extraction.
- **Multiple reports.** Publications from the same study population and period are linked and charted as a single study, retaining the report with the most complete exposure description.

Data charting

Data are charted with a piloted form, built as a structured dataset in R so that it can later be published as the review's open repository. The exposure domains are the core of this review; the shared domains are charted once for studies that also belong to the companion review.

Domain	Fields	Shared with companion review
Identification	Authors, year, journal or series, country, funding	Yes

Domain	Fields	Shared with companion review
Population	Age range, age strata reported, sample size, special groups	Yes
Setting	Country, admin level, urban/rural, Köppen climate, setting (school, home, community, sport)	Yes
Design and analysis	Design, period, statistical model, lag structure, shape of exposure–response, confounders and modifiers	Yes
Metric definition (RQ1)	Metric name and class; input variables; formula or software (e.g. `thermofeel`, `HeatStress`, RayMan); temporal aggregation; time window (24 h, daytime, school hours); threshold type and value; event definition (intensity, duration)	No
Exposure assessment (RQ2)	Data source and product (e.g. station network, ERA5, ERA5-Land, ERA5-HEAT, MERRA-2, satellite LST, loggers); spatial and temporal resolution; assignment unit; indoor, outdoor or both; correction or downscaling	No
Outcome	Domain, specific outcome, instrument or data source, timing relative to exposure	Yes
Comparative performance (RQ3)	Metrics compared; criterion (AIC, QAIC, R^2, cross-validated error, effect size, threshold detection); best-performing metric; whether differences were meaningful	No
Paediatric justification (RQ5)	Child-specific reason given for the metric, verbatim summary	No
Main finding	Direction and form of the association for each metric (adverse, protective, null, non-linear)	Yes

The form is piloted on ten studies by two reviewers; thereafter one reviewer charts and a second verifies all exposure fields and a 20% sample of the remainder. Corresponding authors are contacted once for missing metric definitions.

Appraisal and synthesis

In line with scoping review methodology, studies are not appraised for risk of bias, with one exception: the comparative-performance subset (RQ3), which directly informs the indicator decision.

Appraisal of comparative studies. Each comparative study is assessed with a short bespoke checklist on whether: (i) metrics were derived from the same source data and spatial resolution; (ii) models were otherwise identical (outcome, lags, confounders, functional form); (iii) the comparison criterion was appropriate and stated in advance; (iv) performance was evaluated out of sample; (v) differences were tested or quantified rather than ranked only. Results inform how much weight each comparison receives; no study is excluded on this basis.

Synthesis and presentation.

- Descriptive counts and proportions of metrics by class, data source, temporal aggregation and threshold type (RQ1–RQ2).
- Evidence maps: metric × outcome domain, and metric × region and climate, highlighting Latin America and tropical and Andean settings (RQ4).
- Narrative synthesis of the comparative subset, tabulating, for each study, the metrics compared, the criterion and the best performer, with appraisal results alongside. Vote counting of "best" metrics is reported only descriptively and with its limitations (RQ3).
- Thematic summary of paediatric justifications (RQ5).

No meta-analysis is planned; effect sizes are charted to describe direction and form, not to be pooled.

From synthesis to indicator selection for WP2.1

The review will recommend one primary and one complementary indicator for WP2.1, chosen with criteria fixed in this protocol before any data are seen. Feasibility criteria act as gates; evidence and relevance criteria are scored.

Code	Criterion	Type	Question asked of each candidate metric	Main evidence source
F1	Historical computability	Gate	Can it be computed hourly from ERA5-Land (0.1°) for all schools, within the 07:00–17:00 school window?	Product documentation
F2	Projection computability	Gate	Can it be computed from CMIP6 (bias-adjusted) variables to 2050, and with what added uncertainty (radiation, wind, daily-only resolution)?	Product documentation
F3	Coherence with WP2.2	Gate	Can the same metric be computed from the temperature and relative humidity loggers used in classrooms and outdoors, so that WP2.2 exposure–response curves can be applied to WP2.1 maps?	WP2.2 protocol
E1	Pediatric association evidence	Scored 0–2	Is the metric consistently associated with child or adolescent outcomes, including cognitive outcomes?	RQ1, RQ4

Code	Criterion	Type	Question asked of each candidate metric	Main evidence source
E2	Comparative performance	Scored 0–2	Does it perform at least as well as alternatives in appraised comparative studies in children?	RQ3
E3	Physiological relevance for children	Scored 0–2	Does it represent the heat exchange routes most relevant to children (humidity, radiation, low air movement indoors)?	RQ5, physiology literature
P1	Policy usability	Scored 0–2	Is it used, or easily adopted, by meteorological and health authorities in Chile, Colombia and Peru, with interpretable categories?	National documents, WP1
P2	Comparability	Scored 0–2	Is it widely enough used to compare results with the international literature?	RQ1, RQ4

Procedure.

1. Candidate metrics are all metrics used in at least three included pediatric studies, plus any metric rated favorably in a comparative study.
2. Each candidate is checked against gates F1–F3. A metric failing F1 or F2 is excluded; a metric failing only F3 can be retained as the complementary indicator, with the limitation stated.
3. Two reviewers score the remaining candidates on E1–P2 independently, then agree a joint score with justification.
4. The scores and the comparative evidence are discussed in a structured session with the WP2.1 and WP2.2 leads and an external biometeorologist. The session records the decision and any dissent.
5. The outcome is a primary indicator, a complementary indicator and a sensitivity metric, reported in a decision table published with the review.

Anticipated tension. Air temperature is likely to pass every gate and to score high on comparability, whereas radiation-dependent indices (WBGT, UTCI) may score higher on physiological relevance but fail F3 and carry more uncertainty under F2. CMIP6 outputs are mostly daily, so the school-hours window may only be applicable to historical data unless a diurnal disaggregation method is adopted (we know min happens at night and max, possibly, during school hrs → Assumption though).

Timeline, deviations and dissemination

1. **5 November 2026** — finalize protocol, close decision points, register on OSF; build and test search strings against the seed set.
2. **12 November 2026** — run searches; calibration and title/abstract screening.

3. **20 November 2026** — full-text screening; charting of exposure domains, prioritizing the comparative subset.
4. **30 November 2026** — appraisal of comparative studies; indicator selection session
5. **15 Dec 2026** — completion of charting, evidence maps and manuscript.

Dissemination and data sharing. The manuscript will be submitted open access. The full charting dataset, search strings, R code and decision table will be deposited on Zenodo under a CC BY 4.0 licence, consistent with the project's outputs management plan, and presented as a browsable repository on the EscuelasCool website. A short Spanish-language technical note will summarise the indicator decision for national counterparts.

References

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